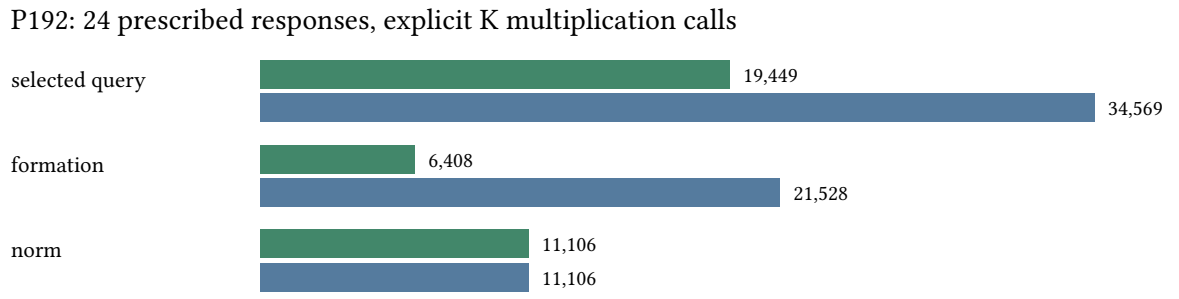
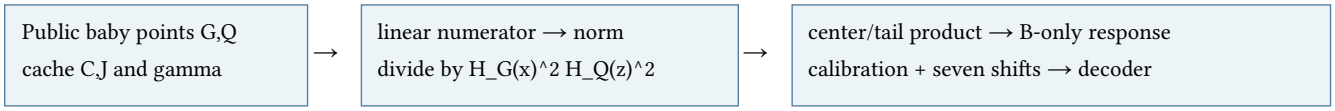
# Fixed coefficient cache for signed pair numerators




Green coefficient cache; blue direct reduced. Shared scale. Setup separate. Source: audit-summary.json and raw P192 rows.




P192 executes prescribed 27-factor windows; the complete decoder panel runs on the small curves.

Round68, 2026-11-09, T-1989. Implemented the9+9 coefficient-polynomial cache from the reduced biquadratic numerator. Once built for a public target, numerator formation uses scalar linear combinations of reduced polynomials, with no query polynomial products. All frozen response controls pass. Scientific standing is unverified producer evidence. [pcc:unverified]

Construction. Expand D_G,N_G,F_G in the giant coordinate x and D_Q,N_Q,F_Q in z. Each coefficient triple has length3. Cache C_ij=FG_i*DQ_j+DG_i*FQ_j-2*NG_i*NQ_j and J_ij=2*(NG_i*DQ_j-DG_i*NQ_j) modulo H_G. Then M0=sum C_ij*x^i*z^j+8*y*w*gamma and M1=sum J_ij*x^i*z^j. The selected block still computes Norm(M0+s*M1)/(H_G(x)^2*H_Q(z)^2). See coefficient-cache-proof.md for explicit triples and correctness. [pcc:unverified]

Formation bound. For S cached coefficient slots and gamma length g, the actual formation stage makes S+g+13 K multiplication calls:2 squares,9 giant monomial weights,2 gamma-scalar calls, then S+g coefficient updates. Since S<=18b and g<=b, this is at most19b+13. No field inverse occurs in this stage. Cache setup is O(b^2) with the current classical polynomial arithmetic and is charged once per changing target. Field norm work and query point arithmetic remain separate. [pcc:unverified]

Universal check. The independent checked-integer polynomial bin now reconstructs M0 and M1 from all18 formal coefficient supports in Z[x,t,z,v,a,b,c], before v=beta(T). It matches the direct reduced expressions and the previous E0/E1 cancellation identities. Correct cache residuals vanish; a changed coefficient gives a nonzero residual. It also retains the wrong-gamma-sign control. The checker shares no elliptic or finite-field arithmetic with the response evaluator. [pcc:unverified]

Primary P192. Every one of72 prescribed degree4 baby blocks uses exactly89 formation multiplications and0 inversions, attaining the bound at b4 with72 cache coefficients and gamma length4. Formation totals6408/0 multiplication/inversion calls versus21528/144 for direct reduced formation. Norm work is unchanged,11106/1008 in each mode, retaining54 unit and18 simple-common-root cases. All24 prescribed27-factor response coefficients preserve the earlier publicQ,mu_G,relation_lambda_G records. This is a prescribed subset, not a complete order-r response evaluation. [pcc:unverified]

Integrated costs and setup. P192 selected queries total19449 multiplications/1317 inversions/168 point-add calls versus34569/1461/168 for direct reduced numerators. Each target's new cache costs1138 multiplications/18 inversions and72 coefficients. Across3 retained targets that adds3414/54 and216 coefficients. Selected queries plus the incremental cache therefore total22863/1371/168 for this panel; common point/algebra/H_Q setup and reference/decoder work are separately recorded. The qualifier also builds the cache in its direct control, records that unused diagnostic overhead, and excludes it from baseline charged setup. These are exact counted panel quantities, with no timing or one-query speedup claim. [pcc:unverified]

Small controls. All381 decoder cases and2694 coefficient queries preserve the prior response values, recovered points and decoder counts. All11382 complete baby blocks match direct paired factors. Direct query receipts preserve Round67. Selected cached/direct totals (multiplication including inverse powers,inversion,point-add calls) are: r7(66220,2800,1400)/(99404,4274,1400); r19(2158609,85475,19296)/(3295513,123267,19296); r199(8109124,188871,10150)/(12338608,261615,10150). New constructor totals by panel are(2932,132,0),(16492,552,0),(34033,741,0); actual cache slots per target are18,34-36,126 respectively. P192 and small counter models stay separate. [pcc:unverified]

Validation and custody. Final native sourcebb566689c756d8370281497f2591cfadf4deba8c and frozen scope precede data. Pre-build review corrected the synthetic-control constructor signature to pass curve coefficients; both source revisions remain. Five science records/two operation receipts reproduce byte-for-byte from a fresh source copy. Altered calibration and P192 coefficient records reject101. No native failure or retry occurred. The inherited library compilation timeout124 remains without retry. Publication preserves earlier artifact-bot manifests and subtree hashes, adding only this round plus claim/ledger/log. No cloud job, generator, role, quota or old freeze lane changed. [pcc:unverified]

Next obligation. With O(b) numerator formation implemented, investigate whether the shared biquadratic family permits batching first norms over giant indices without constructing a dense bivariate resultant surface. Keep signed point transport and every precomputation cost explicit. The measured norm stage now identifies that question directly. [pcc:unverified]

Sources checked. Round67 cancellation-proof.md and sealed outputs; Round64 resultant-proof.md; BFLS arXiv2003.10118 Lemma4.3/section4.16 read in the preceding round. This step extracts fixed coefficients from that retained expression. KB tools/wrapper unavailable. No novelty assertion for basic caching or an imported fast coupled-resultant bound. [pcc:unverified]